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Customer Model

Getting Your Fare Share:

Data-Driven Strategy for NYC Yellow Cab Drivers

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Date: 8/6/2026

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Executive Summary

This report converts observational records of 100,000 New York City yellow cab trips from February 2026, reduced to 62,316 usable observations after cleaning, into operational guidance for drivers facing three recurring decisions that shape shift profitability: which trips yields higher gratuities, where to launch a morning shift, and how long to wait for a fare before relocating. Each question is matched to a model whose structure reflects the shape of its outcome, and inference throughout relies on cluster-robust standard errors at the pickup-zone level to respect the correlation of trips originating in the same location.

Tipping is examined with a two-part Hurdle model estimated on 55,053 credit-card trips, of which 95.5% record a positive tip (*Appendix B, Table B1 - B2*). Because tipping is near universal among players, the decision of whether to tip carries little systematic variation (first-stage pseudo- $R^2 = 0.0086$), and the economically meaningful signal lies in the amount. Tip percentage declines sharply with fare size (Stage 2 coefficient = -0.3105 , $p < 0.001$) and increases with the presence of tolls (coefficient = 0.1401 , $p < 0.001$), a pattern consistent with passengers anchoring to around amounts on larger bills while rewarding the perceived effort of toll and airport routes.

Morning zone selection is framed as a binary success outcome, defined as fare-per-minute at or above the global median of \$1.16, and modelled with logistic regression on 7,862 morning trips (*Appendix C, Table C1*), reaching an AUC of 0.79. Drop-off zone quality (marginal effect = 0.93 , $p < 0.001$), weekend timing (marginal effect = 0.37 , $p < 0.001$) and an earlier start (-0.10 per hour, $p < 0.001$) emerge as the dominant levers, and model-based zone rankings identify JFK Airport (65.9%) and the Upper West Side (47.4%) as the strongest points.

Waiting time between fares is treated as a time-to-event outcome and modelled with a Cox proportional hazards specification (*Appendix D, Table D1*), with the proportional-hazards assumption confirmed by Schoenfeld residual tests for every covariate (*Appendix D, Table D4*) and a Weibull Accelerated Failure Time model retained as a parametric cross-check. Drop-off location is decisive: an outer-borough drop-off lowers the pickup hazard by roughly 80% (hazard ratio = 0.20), equivalent to waits about five times longer, which translates directly into concrete relocation threshold. Collectively, the three analyses indicate that a driver's earnings depend less on in-trip behavior than on the structural choices of where to position, when to start, and where each trip is allowed to end.

1. Introduction

New York City yellow cab drivers operate in one of the world's most competitive urban transport markets, where earnings depend not only on fares but on strategic decisions made before and during each shift. This report addresses three such decisions using observational data on 100,000 yellow cab trips taken in February 2026, sourced from the TLC trip record database and enriched with demographic, weather, and zone-quality variables.

This analysis addresses three questions relevant to driver profitability. Tipping behavior is analyzed through a two-part Hurdle model (Logit + Gamma GLM), separating the decision to tip from the amount given. Morning zone selection is approached as a binary outcome using Logistic Regression, identifying where drivers should begin their shift to maximize trip efficiency. Finally, the risk of extended waiting times between pickups is modeled using a Cox Proportional hazards model, with a Weibull AFT specification retained as a parametric cross-check.

Throughout, actionable recommendations are prioritized. Given the observational nature of the data, findings should be interpreted as associations rather than causal claims, but they nonetheless offer defensible strategic guidance for day-to-day decision-making.

2. Exploratory Data Analysis

2.1 Data Cleaning

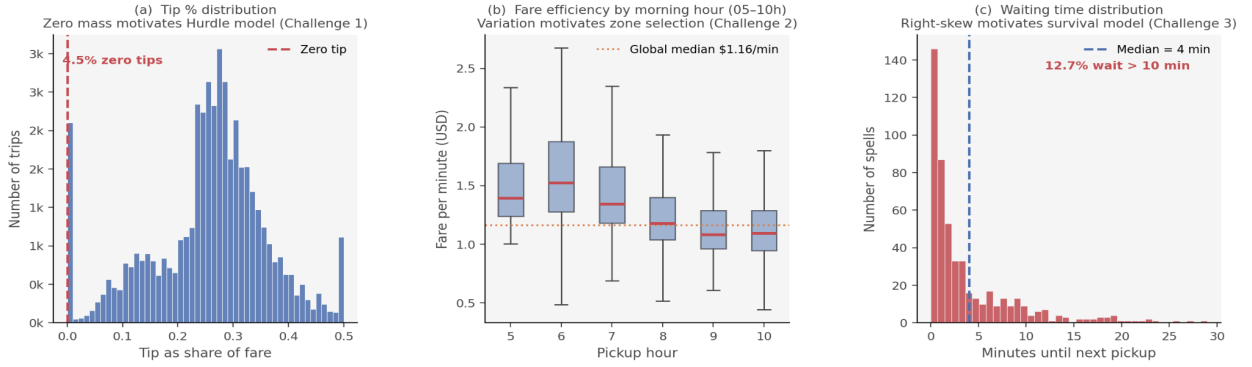
The raw dataset contains 100,000 trips. After merging with TLC taxi zone demographics and hourly weather records, the working dataset spans 47 variables. Cleaning reduced the sample to 62,316 usable observations (a 37.7% reduction), following the steps summarized in Appendix A1.

Missing values in `passenger_count`, `RatecodeID`, `Airport_fee`, and `congestion_surcharge` (each ~30.3%) were entirely attributable to FlexFare trips, which were excluded on substantive grounds above. After exclusion, no missing values remained in the analytic sample. All continuous predictors with right skew were log-transformed where appropriate (`log_fare`, `log_tolls`, `log_distance`). Multicollinearity was assessed via Variance Inflation Factors (VIF) prior to each model; variables exceeding $VIF > 10$ were dropped (`log_distance`, `Airport_fee`, `PU_Borough_Queens` in Challenge 1).

2.2 Data Description

The cleaned sample covers 62,316 trips across all five NYC boroughs in February 2026.

Figure 1. Distributional patterns motivating the three challenge models



Three distributional patterns motivate the analytical approach taken in each challenge, as illustrated in Figure 1. First, 88.3% of trips are paid by credit card while cash tips are not recorded by the meter, so tip analysis is necessarily restricted to card transactions. The tip percentage distribution further reveals a pronounced mass at zero alongside a right-skewed positive tail (*Figure 1a*), a shape that cannot be adequately captured by a single-equation model and directly motivates the two-stage Hurdle specification in Challenge 1. Second, morning trips (05:00–10:00) represent 12.6% of the sample with a 50% success rate across zones, and fare-per-minute varies substantially across pickup hours with earlier hours consistently outperforming later ones (*Figure 1b*), confirming that zone and timing selection at shift start is a meaningful strategic decision. Third, the median waiting time between pickups is 4 minutes, but 16.5% of spells end without a pickup and the distribution is heavily right-skewed with a long tail of extended waits (*Figure 1c*), suggesting that location and timing create structurally different waiting conditions across the city and motivating the survival model approach in Challenge 3. Detailed summary statistics underlying each pattern are reported in Appendix A.

3. Data Interpretation

3.1 Challenge 1 — What Drives Tipping?

Research question: *What determines (a) whether a passenger tips and (b) how large the tip is, as a percentage of the fare?*

3.1.1 Variable Selection

After cleaning, the estimation sample contains 55,053 credit-card trips as cash tips are not recorded by the meter. Of these, 95.5% recorded a positive tip.

The dependent variables are defined as follows. The first-stage outcome is a binary indicator equal to one if a tip was recorded and zero otherwise. The second-stage outcome is the tip as a percentage of the fare, estimated only on the subsample of trips with positive tips. Using

the percentage formulation rather than the absolute tip amount makes the measure comparable across trips of different lengths.

The independent variables capture four groups of factors that drivers can observe before or during a trip: trip economics (log of fare, fare per mile, and log of tolls), temporal conditions (pickup hour and weekend indicator), trip context (passenger count and airport origin), and weather conditions (temperature, a bad-weather dummy, and snowfall). All candidate predictors were checked for multicollinearity using variance inflation factors. After removing redundant variables, all retained regressors have VIF values below 2 (*Appendix A, Table A3*).

3.1.2 Model Strategy

Tipping behaviour consists of two sequential decisions: passengers first decide whether to tip, and conditional on tipping, decide how much to give. A two-part Hurdle model is therefore estimated, with the first stage using a Logit specification for the probability of a positive tip and the second stage using a Gamma regression with a log link for the tip percentage among trips that recorded a tip. The Gamma distribution is chosen because the tip-percentage distribution is strongly right-skewed and bounded at zero. The Hurdle specification is preferred over Tobit because the latter imposes the restrictive assumption that the same latent process governs both the decision to tip and the amount tipped; a formal Cragg test rejects this restriction in favour of the more flexible two-part structure. Discretising the continuous tip percentage into arbitrary bins is also avoided, as it would discard useful variation in the intensive margin. Standard errors are clustered at the pickup-zone level to account for unobserved zone-specific factors that may affect tipping behaviour across trips.

3.1.3 Model Fit

The first-stage logit explains very little variation in the decision to tip (McFadden pseudo- $R^2 \approx 0.01$). This result is expected: 95.5% of credit-card passengers in the sample leave a tip, so there is little systematic variation left to explain once payment method is held constant. The second-stage Gamma regression fits the right-skewed tip-percentage distribution well. A Cragg test strongly favours the hurdle model over a Tobit specification (LR statistic = 8,813, $p < 0.001$), confirming that the two margins of tipping are best modelled separately (*Appendix B, Table B1 - B2*).

3.1.4 Interpretation of Results

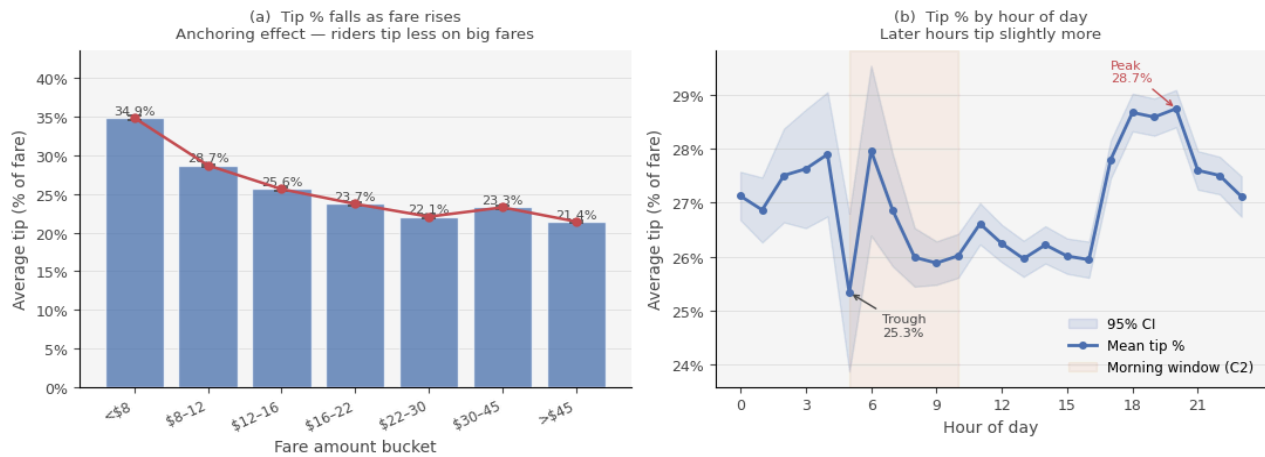
Table 1. What moves the tip (direction and strength of the main drivers)

Driver	Tip a higher %?	Notes for the driver
Higher fare	↓ Lower %	Riders anchor to round amounts → smaller % on big fares

Tolls present (airport/bridge)	↑ Higher %	Strongest positive lever on tip percentage
Later in the day	↑ Slightly higher	Evening riders tip a touch more
Weekend	↓ Slightly lower	Leisure trips tip marginally less
More passengers	↓ Lower (yes/no)	Group trips slightly less likely to tip
Warmer temperature	↑ Slightly higher	Minor comfort effect

Full coefficients, marginal effects and standard errors: Appendix Table B1 - B2.

Figure 2 . Tip percentage patterns by fare size and time of day (credit card tipped trips only, $N = 52,549$)



The dominant pattern is that tip percentage moves inversely with fare size, suggesting passengers mentally anchor to round amounts and tip a smaller share on larger bills (*Figure 2*). The clearest positive lever is the presence of tolls: airport and bridge routes consistently attract higher tip rates, likely because passengers associate these trips with greater effort and distance.

3.1.5 Actionable Insight

➔ **Focus on turnover:** Credit-card passengers can be treated as near-certain tippers. Drivers should therefore focus their effort on trip selection and rapid turnover rather than on trying to “earn” the tip through special service .

➔ **Prioritize toll routes and airport trips:** Airport trips and routes with tolls deliver the highest tip percentages in the sample. Whenever operationally feasible, drivers should prioritise these trips because they combine higher fares with stronger tipping behavior.

→ **Avoid chasing large fares:** Although big fares generate higher absolute tips, they tend to come with smaller tip shares. A consistent flow of standard-length credit-card trips usually produces higher cumulative tips over a shift than occasional high-value outliers.

3.1.6 Robustness and Sensitivity Checks

Splitting by airport versus non-airport shows that in airport trips ($N = 1,790$), only passenger count remains significant ($ME = -0.011$, $p < 0.05$), while the broader set of fare and time variables loses significance, likely reflecting the more standardised nature of airport fares. Among non-airport trips ($N = 53,263$), all four core predictors replicate with consistent signs and significance. Splitting by weekday versus weekend confirms that fare-per-mile, pickup hour, and passenger count are significant in both subsamples, with somewhat larger effects observed on weekends. Results do not hinge on any single segment (*Appendix B, Table B4*).

3.1.7 Model Comparison

The hurdle beats the Tobit on Cragg's test, and beats a single binary logit because it also explains tip size, not just incidence. A multinomial/ordered model in tip "bins" was rejected as it would impose arbitrary cut-offs on a continuous outcome. (*Appendix B, Table B3*)

3.2 Challenge 2 - The Morning Launch

Research question: *Where should a driver begin a morning shift (05:00 - 10:00) to maximise the chance of a "successful" first trip, and what market dynamics explain it?*

3.2.1 Variable Selection

The morning analysis is restricted to trips with pickup hours between 05:00 and 10:00 ($N = 7,862$), representing 12.6% of the cleaned sample. A trip is classified as successful if its fare-per-minute meets or exceeds the global sample median of \$1.16 per minute, a threshold that captures both speed and fare efficiency rather than raw fare alone. This definition is intentionally global rather than zone-relative, ensuring that recommendations are comparable across locations and directly actionable for drivers choosing where to start their shift.

Variables were selected to capture the key dimensions of morning trip profitability. Pickup zone characteristics are represented by population density (PU_Pop_1E) and borough groupings (PU_Borough_grouped_Queens). Dropoff quality (do_zone_quality) reflects the average earnings potential of the zone where a driver is likely to end up after a trip, acknowledging that a good start also depends on where it leads. Time and weather controls include pickup_hour, is_weekend, temperature_2m, bad_weather, and snowfall. All variables passed VIF screening with values below 2.5. (*Appendix A, Table A3*)

3.2.2 Model Strategy

The Binary Logit model rests on two key assumptions. First, observations are assumed to be independent conditional on the covariates included. This assumption is addressed by applying

cluster-robust standard errors at the pickup zone level, accounting for the fact that trips originating from the same zone share unobserved characteristics. Second, the model assumes a linear relationship between predictors and the log-odds of success.

3.2.3 Model Fit

Model fit is assessed through multiple criteria. The Pseudo R^2 of 0.185 indicates meaningful explanatory power relative to a null model. The AUC of 0.79 confirms strong discriminatory ability well above the random baseline of 0.50, and the Brier score of 0.186 indicates well-calibrated predicted probabilities against the baseline of 0.25 (*Appendix C, Table C1*).

3.2.3 Interpretation of Results

Table 2 summarizes the key drivers of morning trip success in plain terms

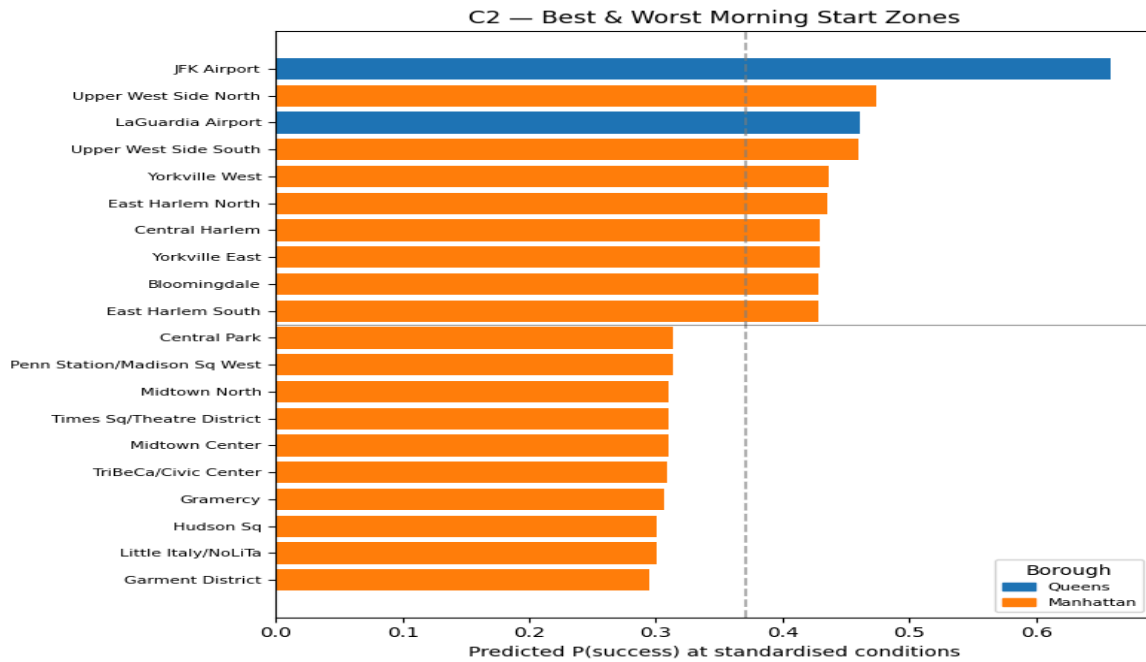
Table 2 . What drives a successful morning trip?

Driver	Effect on success	Notes for the driver
Dropoff zone quality	↑ Much higher	Land in a productive zone compounds earnings across whole shift
Weekend morning	↑ Much higher	Saturday and Sunday mornings are structurally more profitable; stronger demand
Earlier start (per hour)	↑ Higher	Every hour of delay costs roughly 10 percentage points in success probability; starting at 06:00 vs 08:00 makes a real difference
Queens pickup	↑ Higher	Queens outperforms comparable Manhattan zones, driven by concentration of longer airport-bound trips
Denser pickup zone	↑ Slightly higher	More passengers nearby means shorter dead-miles between trips
Warmer temperature	↑ Slightly higher	Greater outdoor mobility on warmer mornings
Airport pickup	No significant effect	Once zone quality and borough are controlled for, being at an airport adds little extra advantage
Bad weather / snowfall	No significant effect	Adverse conditions do not significantly alter morning success

Full coefficients, marginal effects and standard errors: Appendix Table C1

At the zone level, model-based predictions identify JFK Airport (65.9%), Upper West Side North (47.4%), and LaGuardia Airport (46.1%) as the strongest morning starting locations. Penn Station/Madison Square West (31.3%), Midtown Center (31.0%), and Garment District (29.5%) consistently rank at the bottom (*Appendix C, Table C5*)

Figure 3. Best and Worst Morning Start Zones



3.2.4 Actionable Insight

→ **Start early:** A driver starting at 06:00 instead of 08:00 enters the shift with a 20 percentage point structural advantage before a single decision is made.

→ **Choose the starting zone strategically:** Upper West Side North consistently outperforms airport zones as a morning starting point — success rates are comparable but residential zones offer shorter queues and more flexibility to reposition throughout the shifts.

→ **Think about where the trip ends, not just where it starts:** Prioritizing trips that end in high-demand zones, even at an average fare, positions drivers for the next good trip immediately and compounds earnings across the entire shift.

3.2.5 Robustness and Sensitivity Checks

Three checks confirm the main findings are stable. Re-estimating as an LPM shows sign agreement in 9 out of 10 cases, with closely aligned magnitudes (is_weekend: +0.365 Logit vs +0.335 LPM); the sole disagreement on is_airport is insignificant in both models. Varying the success threshold from the global median to the 75th percentile and a composite definition leaves the same five core predictors significant with consistent directions. Splitting into weekday and

weekend subsamples shows all four key predictors remain significant in both groups, with attenuated effect sizes on weekends reflecting the high baseline success rate of 81.9%. (*Appendix C, Tables C3 and C4*).

3.2.6 Model Comparison

Three specifications were considered. The Binary Logit is preferred over the Linear Probability Model, which produces predicted probabilities outside the $[0, 1]$ range and is therefore unsuitable as a primary specification given the balanced outcome distribution (50% success rate). Probit is theoretically equivalent to Logit under standard assumptions and yields near-identical marginal effects in practice; it is not adopted as a separate model as it offers no practical advantage. The LPM is retained as a robustness benchmark, and sign agreement in 9 out of 10 cases confirms that the Logit findings are not a functional-form artefact (*Appendix C, Table C2*).

3.3 Challenge 3 - The Turnover Trap

Research question: *After a drop-off, how long should a driver wait for the next fare before relocating?*

3.3.1 Variable Selection

The survival analysis uses the full cleaned sample ($N = 62,316$), with waiting time defined as the gap between a dropoff and the next recorded pickup. Spells that do not result in a pickup within the observation window are treated as right-censored (16.5% of spells). The event rate is 83.5% and the median waiting time is 4 minutes

The outcome is the waiting from drop-off to the next pickup, treated as a time-to-event with right-censoring for spells that never convert in the window. Predictors are the drop-off borough, time of day, airport, zone population and weather. All variables passed VIF screening with values below 1.65, confirming no multicollinearity concerns (*Appendix A, Table A3*).

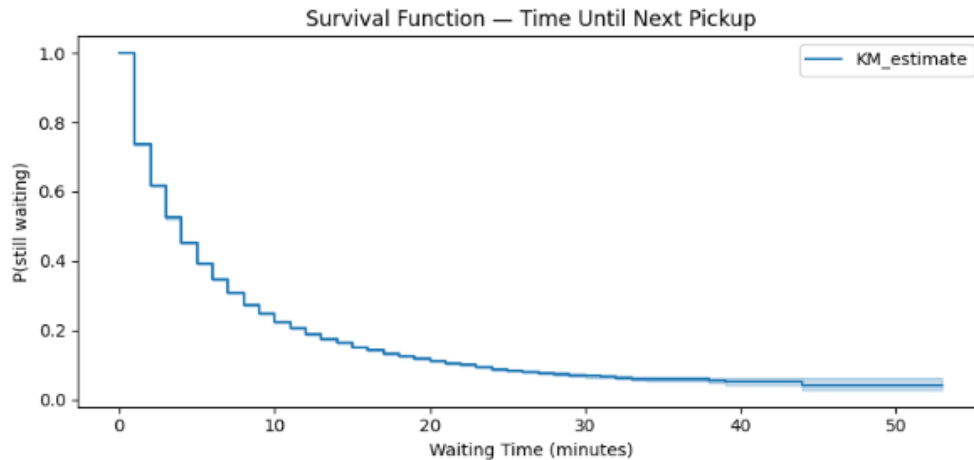
3.3.2 Model Strategy

The hazard of securing a fare is modeled using a Cox proportional hazards model, chosen over fully parametric alternatives such as Weibull because it imposes no assumptions on the shape of the baseline waiting curve. The proportional hazards assumption is verified via Schoenfeld residual tests; all nine covariates return p-values above 0.05, confirming the assumption holds and validating Cox as the primary specification (*Appendix D, Table D4*). A Weibull Accelerated Failure Time model is estimated as a cross-check. As driver IDs are not available in the public dataset, waiting time is proxied by the gap between consecutive pickups within the same zone, and standard errors are clustered at the zone level to account for shared unobserved characteristics.

3.3.3 Model Fit

The model ranks waits better than chance (concordance = 0.58) and the proportional-hazards assumption is not rejected for any variable. The waiting curve itself (*Figure 4*) is steep early then flattens: half of all waits clear within approximately 4 minutes, but a stubborn tail lingers.

Figure 4. Kaplan-Meier survival curve



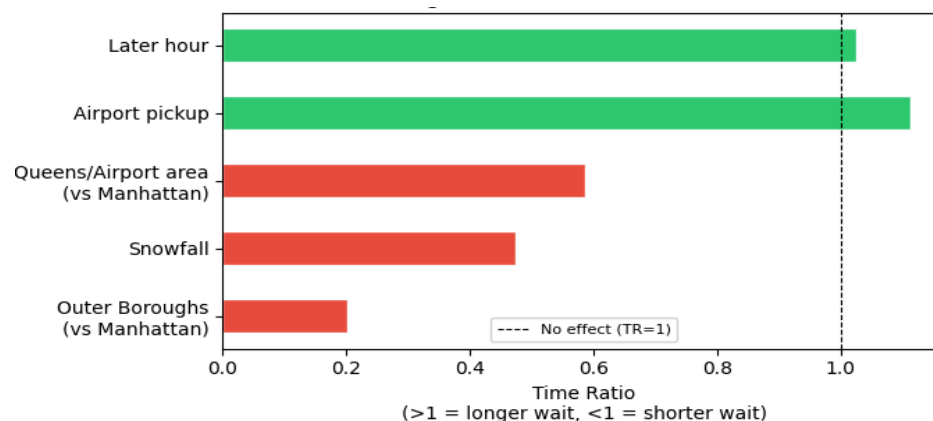
Note: Kaplan-Meier survival curve for time to next pickup: most fares arrive quickly, with a long thin tail of futile waits

3.3.4 Interpretation of Results

Where drop off matters most. Relative to Manhattan, an outer-borough drop-off cuts the chance of a quick pickup by about 80% (roughly five times longer waits) and a Queens drop-off means approximately 73% longer waits. Pickups speed up later in the day, and snow sharply lengthens waits (*Figure 5*).

Figure 5. What speeds up or slows down the next pickup?

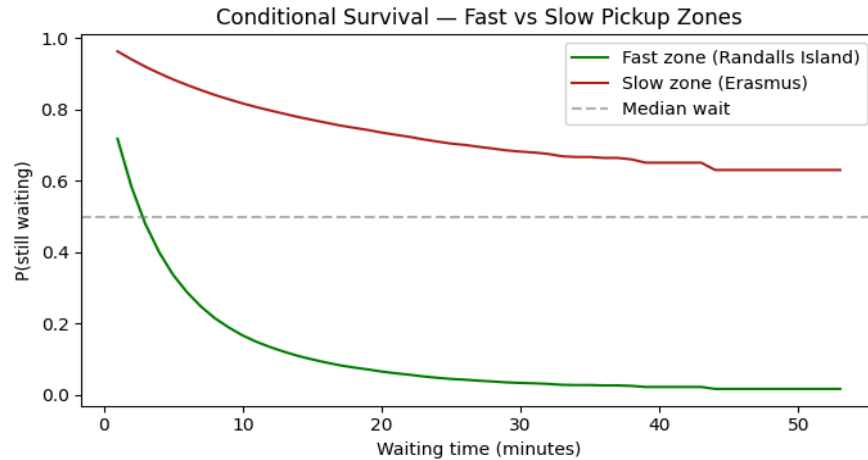
(Significant factors, Cox PH Hazard Ratio)



Note: Cox hazard ratios. Green (>1) = faster next pickup; red (<1) = longer wait. Outer-borough drop-offs and snow are the biggest drags

The conditional-survival curve (Figure 6) makes the rule concrete: in a fast zone the chance of still waiting drops below 50% within approximately 3 minutes, while in a slow zone it stays high for 20+ minutes. Interpret that into a simple wait-or move table.

Figure 6. Conditional Survival - Fast vs Slow Pickup Zones



Note: Conditional survival, fast vs slow zone: how long you are likely to keep waiting. Cross the dashed median early in a slow zone -> relocate

Table 3: When to give up - maximum wait before relocating

Where you dropped off / when	Wait up to	Then
Manhattan core, peak hours	8 - 10 min	Likely a fare soon → remain in position
Airport queue	~10 min	Turnover is fast once in line
Queens	5 – 6 min	Start drifting toward Manhattan
Outer boroughs (Brooklyn/Bronx)	~5 min	Relocate toward Manhattan (empty run is usually faster than continued waiting)
Any zone, heavy snow	Cut the above by half	Demand collapses quickly → reposition early

Full coefficients, marginal effects and standard errors: Appendix Table D2

3.3.5 Actionable Insight

Rule of thumbs: Relocate once the chance of 10+ more minutes of waiting exceeds 50%.

→ **Wait longer in productive zones:** After a drop-off in Manhattan during peak hours or at an airport, drivers can reasonably wait up to approximately 10 minutes, as the instantaneous pick-up rate remains high

→ **Move quickly in outer boroughs:** Pickup rates in outer boroughs are substantially lower than in Manhattan. Relocating toward Manhattan within 5 minutes of a dropoff is typically more profitable than continuing to wait in place.

→ **Shorten all thresholds during heavy snow:** Demand drops sharply across the city in heavy snowfall conditions. All waiting thresholds should be reduced by roughly half and drivers should reposition earlier than usual regardless of location.

3.3.6 Robustness and Sensitivity Checks

Two checks confirm the main findings are stable. First, the Weibull AFT cross-check reproduces every effect direction: outer-borough waits are approximately five times longer than Manhattan (TR = 5.18), Queens waits are 73% longer (TR = 1.73), and snowfall more than doubles expected waiting time (TR = 2.28), consistent with the Cox hazard ratios throughout. Second, trimming the sample to spells of 72 minutes or less (N = 62,191) leaves all hazard ratios stable to two decimal places, confirming that results are not driven by extreme waiting spells (*Appendix D, Tables D4 and D5*).

3.3.7 Model Comparison

Cox (semiparametric) and Weibull (parametric) agree, which is reassuring. And we lead with Cox because it imposes no shape on the waiting curve, and keep Weibull only to confirm robustness (not to fix any violation, since none exists).

4. Overall Discussion and Critical Reflection

A central contribution of this report emerges only when the three challenges are read together rather than in isolation. To make the comparison concrete, each pickup zone was scored on three standardised dimensions drawn from fitted models, namely its mean predicted tip percentage from Challenge 1, its model-based probability of a successful morning trip from Challenge 2, and its mean predicted partial hazard from Challenge 3, where a higher hazard indicates that the next pickup arrives sooner. Each dimension was converted to a z-score and averaged into a composite measure of zone attractiveness, with the spread between a zone's best and worst dimension serving as an explicit flag for trade-offs. The exercise reveals that the zones a driver should prefer depend entirely on which margin of profitability is being optimised, and that no location is uniformly superior.

The composite ranking is headed by residential Manhattan zones: Upper West Side North (composite = 0.53), Upper West Side South (0.52) and Yorkville West (0.51). These zones

combine strong tipping performance (z-scores 0.80–1.02) with high earning efficiency (z-scores 1.00–1.59), yet sit below average on pickup speed (z-scores -0.50 to -0.78). The pattern shows that the most lucrative zones per trip are also those where drivers wait longest between fares; attractiveness therefore rests on trip value rather than rapid turnover.

The opposite profile characterises the central business district. Midtown Center, Central Park and Midtown North record the fastest pickups (speed z-scores up to 1.34) but distinctly weak earning efficiency (success z-scores near -0.95), describing a high-churn environment of frequent yet individually modest trips.

The starkest result concerns the airports. Once tipping and turnover are layered onto earning efficiency, JFK Airport emerges as the largest trade-off zone in the sample (composite = -1.46 , spread = 9.41). It pairs the highest earning-success score (4.46) with the lowest tipping score (-4.95) and a deeply negative pickup-speed score (-3.90). LaGuardia exhibits the same tension (composite = -1.78 , spread = 5.65). This finding does not contradict Challenge 2; it qualifies it. The morning model correctly identifies airports as strong launch points for the first trip, yet the integrated view shows they are high-commitment positions offering poor gratuities and slow re-entry into the fare stream. Consequently, the synthesis vindicates the earlier recommendation to favour the Upper West Side over the airport as a default starting location: comparable first-trip success is delivered with materially better tipping and greater operational flexibility.

The three models were deliberately assigned distinct evaluative standards. Challenge 1 is inferential; a Cragg likelihood-ratio test strongly favours the two-part specification over a single-mechanism Tobit (LR = 8,813, df = 10, $p < 0.001$), consistent with hurdle-model applications such as Bayart et al. (2018). Challenge 2 is predictive and is therefore judged on discrimination and calibration (AUC = 0.79, Brier score = 0.186). Challenge 3 blends explanatory hazard ratios with operational relocation thresholds obtained from the conditional survival function, echoing dynamic customer-behavior modelling in Hu et al. (2021).

Several limitations qualify the conclusions. The data are observational, so every coefficient represents an association rather than a causal effect. Recommendations should be read as defensible operating heuristics. Tipping analysis is confined to credit-card trips (88.3% of the sample) because cash tips are not metered. The near-universal tipping rate (95.5%) renders the participation margin almost degenerate, so substantive findings rest on the amount stage. Waiting-time analysis relies on consecutive within-zone pickup gaps as a proxy for idle time because driver identifiers are absent from the public data. Finally, the sample covers a single winter month, rendering the pronounced snowfall effects period-specific. The composite itself is a descriptive aggregation of model predictions rather than inferential statistics.

Robustness checks nevertheless support the signals feeding the synthesis. Tipping effects retain sign and significance when the sample is split by airport status and by weekdays versus

weekend. The morning logit agrees with a linear-probability benchmark on nine of ten coefficients. Core predictors survive alternative success thresholds and weekday/weekend subsamples. Waiting-time conclusions are reproduced in direction by the Weibull specification, remain stable under the proportional-hazards diagnostic, and hold after trimming extreme spells (*Appendix Table B4, C2-C4 and D3-D5*). Because the composite ranking rests on individually robust predictions, the integrated trade-off structure can be advanced with reasonable confidence for strategic advice to drivers.

5. Conclusion

This analysis distills into three pieces of advice that a driver can act on without access to the underlying models. First, in the credit-card market a tip is close to certain, so effort is better spent on turnover and on selecting toll and airport routes, which combine higher fares with strongest tip percentages, than on chasing large fares whose gratuities shrink as a share of the bill. Second, the morning shift should begin early and be anchored to zones that lead to productive drop-offs rather than to the most visible pickup spots, since starting at 6:00 rather than 08:00 confers roughly a twenty percentage point advantage in the probability of a successful first trip before any further decision is made. Third, waiting time should be calibrated to location, with drivers reasonably holding position for up to about ten minutes in Manhattan or an airport queue, relocating toward Manhattan roughly five minutes after an outer-borough drop-off, and halving every threshold during heavy snow when citywide demand collapses.

The broader value of the analysis is that it reframes a set of intuitive driver habits as quantified, testable relationships, and in doing so separates the decisions that genuinely move earnings, namely positioning, timing and trip selection, from those that do not. Because the findings rest on associations in a single month of observational data, they are offered as evidence-based operating heuristics rather than precise guarantees, but their internal consistency across three independent modelling approaches gives reasonable confidence that the recommended behaviors would on average, improve a driver's productivity over a shift.

Appendices

A. Data Cleaning & Exploratory Data Analysis

Table A1. Data Cleaning Step

Filter	Rows removed	Rationale
Payment type: FlexFare (0), No Charge (3), Dispute (4)	~31,761	Non-standard transactions; tips and fares are not comparable
RatecodeID = 99 (unknown)	~2,971	Rate structure unknown; fares not interpretable
Fare ≤ 0 or total amount ≤ 0	~826 / 783	Likely refunds or data errors
Trip distance = 0	~3,659	No movement recorded; trip did not occur
Trip duration ≤ 0 or > 180 min	~1,238	Invalid or extreme outliers
Passenger count = 0	~350	No passenger present
Negative tolls or tip amounts	~62	Data entry errors
Total removed	37,684 (37.7%)	

Table A2 Full Data Description

Variable	Description	Type	Mean Share	/	Std Dev	Min	Median	Max
tip_yes	1 if tip given (CC trips only)	Binary	95.5%*	—	0	1	1	
tip_pct	Tip as share of fare amount	Continuous	0.272*	—	0.0001	—	—	
trip_success	1 if fare/min $\geq$ \$1.16 (morning subset)	Binary	50.0%**	—	0	1	1	
wait_duration	Minutes between consecutive pickups	Continuous	6.69	6.15	1	4.0	72	

event_observed	1 if pickup occurred; 0 if censored	Binary	83.5%	—	0	1	1
fare_amount	Metered fare (USD)	Continuous	21.62	18.28	2.50	16.30	834.60
trip_distance	Trip distance (miles)	Continuous	—	—	0.01	1.80	—
log_fare	Natural log of fare amount	Continuous	—	—	—	—	—
fare_per_mile	Fare divided by trip distance (USD/mile)	Continuous	—	—	—	—	—
log_tolls	$\log(1 + \text{tolls amount})$, log1p-transformed	Continuous	—	—	0	0	—
passenger_count	Number of passengers (1–6)	Discrete	1.23	0.64	1	1	6
do_zone_quality	Avg fare/min of dropoff zone (USD/min)	Continuous	—	—	0.00	—	2.67
congestion_surge	Congestion surcharge per trip (USD)	Continuous	—	—	0	—	—
payment_type	Credit card (1) / Cash (2)	Binary	CC: 88.3%	—	—	—	—
pickup_hour	Hour of pickup (0–23)	Discrete	—	—	0	—	23
is_weekend	1 if Saturday or Sunday	Binary	27.4%	—	0	0	1
is_morning	1 if pickup between 05:00–10:00	Binary	12.6%	—	0	0	1
temperature_2m	Air temperature at pickup time (°C)	Continuous	—	—	—	—	—
bad_weather	1 if precipitation or high wind	Binary	8.6%	—	0	0	1
snowfall	Snowfall amount at pickup time (cm)	Continuous	—	—	0	0	—

is_airport	1 if pickup at JFK or LaGuardia	Binary	3.3%	—	0	0	1
PU_Pop_1E	Residential population of pickup zone	Continuous	—	—	—	—	—
PU_Borough_grouped_Queens	1 if pickup in Queens	Binary	—	—	0	0	1
PU_Borough_grouped_Outer_Boroughs	1 if pickup in Bronx, Brooklyn, or Staten Island	Binary	—	—	0	0	1
PULocationID	TLC taxi zone ID (zone-level control)	Discrete	—	—	—	—	—

**Restricted to credit card trips (N = 55,053). **Restricted to morning trips 05:00–10:00 (N = 7,862).*

Table A3 Multicollinearity Check (after dropping collinear variables)

Variable	Challenge 1	Challenge 2	Challenge 3
log_tolls	1.67	-	-
log_fare	1.65	-	-
is_airport	1.51	1.26	1.26
fare_per_mile	1.24	-	-
bad_weather	1.23	1.55	1.23
snowfall	1.19	1.44	1.19
temperature_2m	1.10	1.09	1.10
is_weekend	1.07	1.07	1.04
pickup_hour	1.07	1.04	1.07
passenger_count	1.01	-	-
PU_Borough_grouped_Queens	-	2.18	1.61

congestion_surcharge	-	1.85	-
do_zone_quality	-	1.44	-
PU_Pop_1E	-	1.35	1.36
PU_Borough_grouped_Outer Boroughs	-	-	1.00

Note: All retained variables have VIF < 2.5. Variables dropped from C1: log_distance (VIF = 28.17), PU_Borough_grouped_Queens (VIF = 14.73), Airport_fee (VIF = 13.54), trip_duration_min (VIF = 5.76). Weather variables verified separately in notebook environment with real weather data

B. Challenge 1: What Drives Tipping?

Table B1. Stage 1 — Logit regression (tip incidence)

Dependent variable: tip_yes (binary). N = 55,053 credit card trips. Pseudo R² = 0.0086. Cluster-robust standard errors at pickup zone level.

Variable	Coef.	Std. Err.	z	P> z	Marg. Effect
Intercept	3.4683	0.216	16.05	0.000	—
log_fare	-0.0807	0.065	-1.24	0.217	-0.0035
fare_per_mile	-0.0359	0.007	-4.96	0.000***	-0.0016***
pickup_hour	0.0252	0.004	7.15	0.000***	0.0011***
is_weekend	-0.3268	0.055	-5.98	0.000***	-0.0141***
is_airport	0.0646	0.185	0.35	0.727	0.0028
log_tolls	-0.0139	0.098	-0.14	0.888	-0.0006
temperature_2m	0.0004	0.005	0.10	0.921	0.0000
bad_weather	-0.0073	0.054	-0.13	0.894	-0.0003
snowfall	-0.3825	0.271	-1.41	0.158	-0.0165
passenger_count	-0.1111	0.030	-3.68	0.000***	-0.0048***

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Marginal effects (ME) computed at sample means

Table B2. Stage 2 — Gamma GLM (tip percentage | tipped)

Dependent variable: tip_pct (continuous). N = 52,549 tipped trips. AIC = -91,707.5. Mean tip_pct = 0.272.

Variable	Coef.	P> z	Marg. Effect
log_fare	-0.3105	0.000***	-0.0845***
fare_per_mile	0.0015	0.104	0.0004
pickup_hour	0.0024	0.000***	0.0007***
is_weekend	-0.0215	0.000***	-0.0059***
is_airport	-0.0038	0.898	-0.0010
log_tolls	0.1401	0.000***	0.0381***
temperature_2m	0.0009	0.014**	0.0002**
bad_weather	-0.0012	0.864	-0.0003
snowfall	0.0461	0.124	0.0125
passenger_count	0.0013	0.638	0.0004

*** $p < 0.01$, ** $p < 0.05$. Log link function. Marginal effects expressed in percentage-point units of tip_pct.

Table B3. Cragg likelihood ratio test - Hurdle vs. Tobit

Statistic	Value
LL Hurdle (joint)	35,768.10 (k = 22)
LL Tobit	31,361.66 (k = 12)
LR statistic	8,812.88
Degrees of freedom	10
p-value	< 0.001
Decision	Hurdle preferred over Tobit

H0: Single-mechanism Tobit is sufficient. Rejection favors the two-part Hurdle specification.

Table B4. Robustness checks — Challenge 1

Variable	Airport (N=1,790)	p	Non-Airport (N=53,263)	p
Stage 1 — Logit marginal effects (significant predictors only)				
passenger_count	-0.0109*	0.048	-0.0045***	0.001
fare_per_mile	n.s.	-	-0.0016***	0.000
pickup_hour	n.s.	-	0.0011***	0.000
is_weekend	n.s.	-	-0.0149***	0.000
Weekday (N=39,982) vs Weekend (N=15,071)				
fare_per_mile	-0.0010***	0.000	-0.0036***	0.000
pickup_hour	0.0006***	0.001	0.0020***	0.000
passenger_count	-0.0040***	0.007	-0.0060*	0.014

Stage 1 marginal effects shown for significant predictors only. n.s. = not statistically significant at 5% level.

C - Challenge 2: The Morning Launch

Table C1. Binary Logit — full regression output (morning trips)

Dependent variable: trip_success. N = 7,862 morning trips (05:00–10:00). Pseudo R² = 0.185. AUC = 0.79. Brier score = 0.186. Cluster-robust SEs at pickup zone level.

Variable	Coef.	Std. Err.	z	P> z	Marg. Effect
Intercept	-1.1929	0.625	-1.91	0.056	—
pickup_hour	-0.5262	0.040	-13.00	0.000***	-0.1017***
is_weekend	1.8887	0.121	15.59	0.000***	0.3652***
is_airport	-0.0716	0.239	-0.30	0.765	-0.0139
congestion_surcharge	-0.0301	0.077	-0.39	0.694	-0.0058

PU_Pop_1E	0.0063	0.002	3.40	0.001***	0.0012***
do_zone_quality	4.8130	0.600	8.02	0.000***	0.9306***
bad_weather	-0.1223	0.110	-1.11	0.267	-0.0236
temperature_2m	0.0199	0.006	3.08	0.002**	0.0039**
snowfall	-0.3334	0.385	-0.87	0.387	-0.0645
PU_Borough_grouped_Queens	0.7478	0.289	2.59	0.010***	0.1446***

*** $p < 0.01$, ** $p < 0.05$. Marginal effects computed at sample means.

Table C2. Robustness - Logit vs. LPM comparison

Variable	Logit ME	LPM Coef.	Logit p	Same sign?
pickup_hour	-0.1017	-0.0932	0.000***	Yes
is_weekend	0.3652	0.3351	0.000***	Yes
is_airport	-0.0139	0.0497	0.764	No
congestion_surcharge	-0.0058	-0.0126	0.694	Yes
PU_Pop_1E	0.0012	0.0015	0.001***	Yes
do_zone_quality	0.9306	0.5041	0.000***	Yes
bad_weather	-0.0236	-0.0085	0.265	Yes
temperature_2m	0.0039	0.0032	0.002**	Yes
snowfall	-0.0645	-0.0879	0.389	Yes
PU_Borough_grouped_Queens	0.1446	0.1176	0.009***	Yes

LPM estimated with HC3 heteroskedasticity-robust standard errors. Sign agreement: 9/10 predictors. Sole disagreement on *is_airport* is insignificant in both models.

Table C3. Sensitivity — alternative success thresholds

Variable	Main (median)	p75 threshold	Composite	Consistent?
N (successes)	3,931	1,966	893	
Pseudo R ²	0.185	0.178	0.609	
pickup_hour	-0.1017***	Sig.***	Sig.***	Yes
is_weekend	0.3652***	Sig.***	Sig.***	Yes
PU_Pop_1E	0.0012***	Sig.***	Sig.***	Yes
do_zone_quality	0.9306***	Sig.***	Sig.***	Yes
PU_Borough_grouped_Queens	0.1446***	Sig.***	Sig.***	Yes

Main: fare-per-minute $\geq$ global median (\$1.16/min). p75: fare-per-minute $\geq$ 75th percentile. Composite: above-median fpm AND high-quality dropoff zone. 'Sig.' indicates significance at $p < 0.05$ with consistent direction.

Table C4. Robustness — weekday vs. weekend subsamples

Variable	Weekday (N=6,424)	p	Weekend (N=1,438)	p
Success rate	42.9%	—	81.9%	—
pickup_hour	Sig.***	0.000	Sig.***	0.000
do_zone_quality	Sig.***	0.000	Sig.***	0.000
PU_Pop_1E	Sig.***	0.000	Sig.***	0.001
PU_Borough_grouped_Queens	Sig.**	0.010	Sig.**	0.021
is_weekend	— (not applicable)	—	— (not applicable)	—

Four core predictors remain significant in both subsamples. Attenuated weekend effect sizes reflect ceiling compression at 81.9% baseline success rate.

Table C5. Zone-level predicted success probabilities - full ranking

Zone	Borough	N trips	Predicted P(success)
Top 10 zones — highest predicted success			

JFK Airport	Queens	435	0.659
Upper West Side North	Manhattan	251	0.474
LaGuardia Airport	Queens	134	0.461
Upper West Side South	Manhattan	235	0.460
Yorkville West	Manhattan	241	0.436
East Harlem North	Manhattan	23	0.435
Central Harlem	Manhattan	38	0.429
Yorkville East	Manhattan	235	0.429
Bloomingdale	Manhattan	36	0.428
East Harlem South	Manhattan	78	0.428
Bottom 10 zones — lowest predicted success			
Garment District	Manhattan	160	0.295
Little Italy/NoLiTa	Manhattan	21	0.301
Hudson Sq	Manhattan	27	0.301
Gramercy	Manhattan	189	0.307
TriBeCa/Civic Center	Manhattan	76	0.308
Midtown Center	Manhattan	256	0.310
Times Sq/Theatre District	Manhattan	179	0.310
Midtown North	Manhattan	153	0.310
Penn Station/Madison Sq West	Manhattan	418	0.313
Central Park	Manhattan	97	0.313

Predicted probabilities from Binary Logit model, conditional on all covariates at sample means. Raw success rate shown for comparison. Zones with $N < 20$ trips excluded.

D - Challenge 3: Turnover Trap

Table D1. Cox Proportional Hazards model — full output (primary model)

$N = 62,316$. Events observed = 52,053 (83.5%). Concordance = 0.58. Partial AIC = 1,071,534.88. Cluster-robust SEs at pickup zone level.

Variable	Coef.	HR	Std. Err.	z	P> z	95% CI (HR)
pickup_hour	0.03	1.03	0.00	7.35	0.000***	[1.02, 1.03]
is_weekend	-0.04	0.96	0.04	-1.01	0.310	[0.88, 1.04]
is_airport	0.11	1.11	0.05	2.37	0.018**	[1.02, 1.22]
PU_Pop_1E	-0.00	1.00	0.00	-0.56	0.580	[0.99, 1.00]
bad_weather	0.03	1.03	0.02	1.75	0.080*	[1.00, 1.07]
temperature_2m	-0.00	1.00	0.00	-0.71	0.480	[1.00, 1.00]
snowfall	-0.74	0.48	0.08	-9.10	0.000***	[0.40, 0.56]
PU_Borough_grouped_Queens	-0.54	0.59	0.20	-2.71	0.007***	[0.40, 0.86]
PU_Borough_grouped_Outer Boroughs	-1.60	0.20	0.13	-12.53	0.000***	[0.16, 0.26]

HR = Hazard Ratio ($\exp(\text{coef})$). $HR > 1$ = faster pickup (shorter wait). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table D2. Weibull AFT model — full output

$N = 62,316$. Events observed = 52,053. Log-likelihood = -150,167.76. AIC = 300,359.53. Concordance = 0.60.

Variable	Coef.	Time Ratio	Std. Err.	z	95% CI (TR)
PU_Borough_grouped_Outer Boroughs	1.65	5.18***	0.11	14.31	[4.14, 6.49]
PU_Borough_grouped_Queens	0.55	1.73***	0.02	27.22	[1.66, 1.80]
snowfall	0.83	2.28***	0.08	10.53	[1.96, 2.66]
is_weekend	0.05	1.05***	0.01	4.60	[1.03, 1.07]

PU_Pop_1E	0.00	1.00***	0.00	15.11	[1.00, 1.00]
bad_weather	-0.04	0.96*	0.02	-2.03	[0.93, 1.00]
is_airport	-0.12	0.89***	0.03	-4.24	[0.84, 0.94]
pickup_hour	-0.03	0.97***	0.00	-37.00	[0.97, 0.97]
temperature_2m	0.00	1.00	0.00	1.10	[1.00, 1.00]

*Time Ratio (TR) = $\exp(\text{coef})$ in AFT parameterization. $TR > 1$ = longer wait time; $TR < 1$ = shorter wait. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table D3. Weibull AFT — time ratios (robustness cross-check)

Covariate	Time Ratio	Interpretation	Sig.
PU_Borough = Outer Boroughs	5.182	wait ~418% longer	***
snowfall	2.284	wait ~128% longer	***
PU_Borough = Queens	1.730	wait ~73% longer	***
is_weekend	1.048	wait ~4.8% longer	***
PU_Pop_1E	1.003	wait ~0.3% longer	***
bad_weather	0.964	wait ~3.6% shorter	**
pickup_hour	0.971	wait ~2.9% shorter	***
is_airport	0.887	wait ~11.3% shorter	***

*Log-likelihood = -150,167.8. Time ratio > 1 = longer expected wait. Signs and directions match the Cox hazard ratios, so conclusions are robust to model choice. *** $p < 0.01$, ** $p < 0.05$*

Table D4. Schoenfeld residual test — proportional hazards assumption

Variable	Test statistic	p-value	PH holds?
PU_Borough_grouped_OuterBoroughs	1.524	0.217	Yes
PU_Borough_grouped_Queens	1.936	0.164	Yes

PU_Pop_1E	1.037	0.308	Yes
bad_weather	0.288	0.591	Yes
is_airport	0.006	0.937	Yes
is_weekend	1.303	0.254	Yes
pickup_hour	0.438	0.508	Yes
snowfall	0.481	0.488	Yes
temperature_2m	0.744	0.388	Yes

H0: proportional hazards assumption holds. All p-values > 0.05 — PH assumption not rejected for any covariate. Cox PH therefore valid as primary specification.

Table D5. Sensitivity — trimmed sample (spells ≤ 72 minutes)

Variable	HR — Full sample	HR — Trimmed (≤72 min)	Difference
PU_Borough_grouped_OuterBoroughs	0.203	0.225	Stable
PU_Borough_grouped_Queens	0.586	0.603	Stable
snowfall	0.475	0.473	Stable
is_airport	1.114	1.114	Stable
pickup_hour	1.026	1.026	Stable
bad_weather	1.031	1.032	Stable
is_weekend	0.959	0.958	Stable
temperature_2m	0.999	0.999	Stable
PU_Pop_1E	0.998	0.997	Stable

Hazard ratios from Cox PH model. Full sample N = 62,316; trimmed sample N = 62,191 (removes 125 extreme spells). All HRs remain stable, confirming results are not driven by outlier waiting spells.

E - Cross-challenge zone synthesis

Table E1. Top 10 all-round zones (high composites)

Zone	Borough	z_tip	z_success	z_speed	Composite	Spread
Upper West Side North	Manhattan	0.80	1.59	-0.78	0.53	2.37
Upper West Side South	Manhattan	0.83	1.38	-0.66	0.52	2.04
Yorkville West	Manhattan	1.02	1.00	-0.50	0.51	1.52
Lincoln Square West	Manhattan	1.10	0.63	-0.26	0.49	1.36
East Harlem South	Manhattan	0.66	0.87	-0.13	0.47	1.00
Manhattan Valley	Manhattan	0.73	0.58	0.08	0.46	0.65
Upper East Side South	Manhattan	0.98	0.09	0.31	0.46	0.89
Lenox Hill West	Manhattan	1.06	0.68	-0.38	0.45	1.43
Lincoln Square East	Manhattan	0.66	0.42	0.22	0.43	0.44
East Harlem North	Manhattan	0.35	0.99	-0.07	0.42	1.07

Table E2. Bottom 10 zones (low composites)

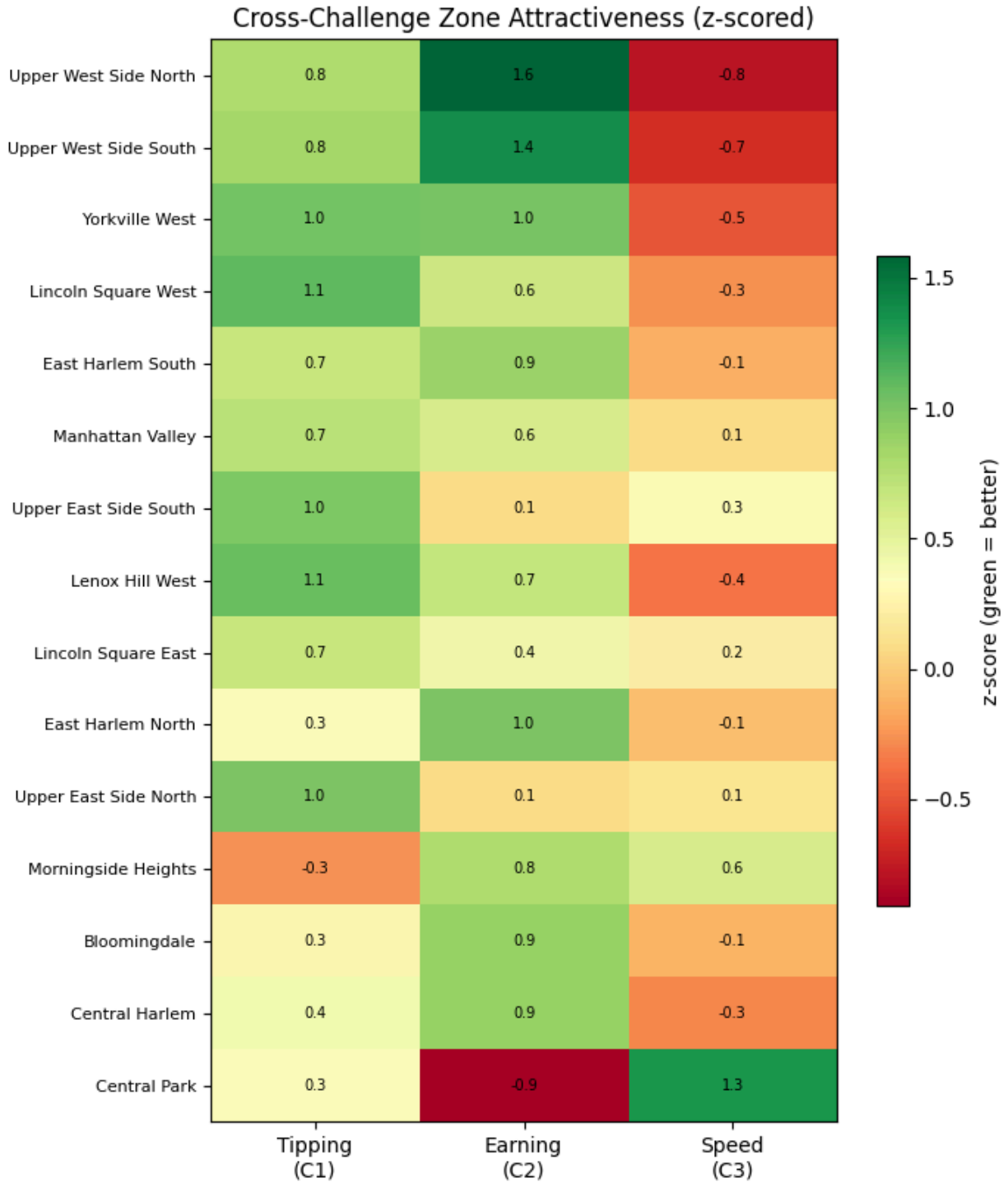
Zone	Borough	z_tip	z_success	z_speed	Composite	Spread
TriBeCa/Civic Center	Manhattan	-0.68	-0.98	0.76	-0.30	1.74
Meatpacking/ West Villa West	Manhattan	-0.35	-0.74	0.19	-0.30	0.93
Greenwich Village South	Manhattan	-0.27	-0.75	-0.05	-0.36	0.71
Battery Park City	Manhattan	-1.46	-0.19	0.57	-0.36	2.04
World Trade Centre	Manhattan	-1.40	-0.19	0.47	-0.37	1.87
Little Italy/NoLiTa	Manhattan	-0.36	-1.09	0.28	-0.39	1.37

Financial District North	Manhattan	-1.55	-0.19	0.10	-0.55	1.65
Lower East Side	Manhattan	-0.71	-0.44	-1.01	-0.72	0.57
JFK Airport	Queens	-4.95	4.46	-3.90	-1.46	9.41
LaGuardia Airport	Queens	-2.45	1.38	-4.27	-1.78	5.65

Table E3. Largest trade-off zones (high spreads)

Zone	Borough	z_tip	z_success	z_speed	Composite	Spread
JFK Airport	Queens	-4.95	4.46	-3.90	-1.46	9.41
LaGuardia Airport	Queens	-2.45	1.38	-4.27	-1.78	5.65
Upper West Side North	Manhattan	0.80	1.59	-0.78	0.53	2.37
Midtown Center	Manhattan	0.12	-0.96	1.29	0.15	2.25
Central Park	Manhattan	0.34	-0.91	1.34	0.26	2.25
Midtown North	Manhattan	0.36	-0.96	1.18	0.20	2.14
Upper West Side South	Manhattan	0.83	1.38	-0.66	0.52	2.04
Battery Park City	Manhattan	-1.46	-0.19	0.57	-0.36	2.04

Figure E1. Cross-challenge zone attractiveness



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